RAMDEOBABA UNIVERSITY, NAGPUR

SCHOOL OF COMPUTER SCIENCE & ENGINEERING

PROJECT SYNOPSIS

Session: 2026–27 | Semester: V | Programme: B.Tech. CSE

ENTRIX

“AI-POWERED INTEGRATED ENTERPRISE RESOURCE PLANNING PLATFORM”

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Date: 16 / 09 / 2026

1. Background and Motivation

ENTRIX is a cloud-based, AI-powered Enterprise Resource Planning (ERP) platform proposed for small and medium-sized organizations. The project responds to fragmented business operations in which Human Resources, Finance, Inventory, Procurement and Project teams may depend on separate spreadsheets or applications. Such fragmentation can scatter business data, require repeated information entry, increase inconsistencies, slow approvals and make management reporting difficult. The proposed platform centralizes organizational information and connects departmental workflows through a common system. It combines operational ERP modules with role-based security, dashboards, audit logging, real-time notifications and AI-assisted decision support. The project also provides an integrated academic implementation of full-stack development, database management, cloud computing, cybersecurity, software architecture and machine learning.

2. Problem Statement

Organizations often manage HR, finance, inventory, procurement and projects through disconnected systems. This leads to scattered data, duplicate entry, manual errors, slow approval processes, weak coordination and limited real-time visibility. Reports may require additional manual effort, while conventional systems may provide limited intelligent decision support. ENTRIX addresses these issues through a centralized ERP platform with shared organizational data, workflow automation, secure role-based access and selected AI features for forecasting, anomaly detection and management insights.

3. Objectives

- Develop an integrated ERP platform for organizational management.
- Centralize business information in a secure relational database.
- Automate repetitive business workflows and approval processes.
- Integrate HR, Finance, Inventory, Procurement and Project Management.
- Provide real-time dashboards, business reports, notifications and audit trails.
- Implement AI-powered forecasting, anomaly detection and business insights.

4. Literature Review and Research Gap

Research on ERP systems in SMEs identifies implementation characteristics and research gaps specific to smaller organizations [1]. ERP integration literature classifies integration into technical, business and socio-organizational dimensions, highlighting the importance of coordinated information and processes [2]. Cloud-ERP studies examine implementation factors and challenges for SMEs, including technological, management and environmental considerations [3], [4]. Research on cloud-ERP also indicates that some ERP lifecycle stages have stronger evidence than others [4]. For intelligent functions, anomaly-detection research reviews supervised, semi-supervised and unsupervised techniques for financial applications [5], while machine-learning demand forecasting has been investigated for inventory management [6]. Cloud ERP implementation research additionally identifies organizational, environmental, technological and individual factors affecting deployment [7]. These studies provide foundations for the proposed platform.

Research Gap: The reviewed literature addresses ERP integration, SME/cloud implementation and individual AI applications largely as separate research areas. ENTRIX proposes an academic prototype that combines shared ERP data, departmental workflows, RBAC, audit trails, real-time updates and selected AI services within one modular platform.

Sr.No.	Author(s) & Year	Title / Focus; Methodology / Key Contribution	Limitation / Gap
1	Haddara & Zach, 2011	ERP Systems in SMEs: A Literature Review. Reviews 77 articles and organizes findings around ERP lifecycle phases.	Focuses on ERP research in SMEs; newer cloud/AI combinations are outside its period.
2	Aguirre Mayorga et al., 2018	ERP and IS integration. Systematic review using technical, business and socio-organizational integration dimensions.	Identifies research opportunities rather than implementing a unified platform.
3	Alsharari, Al-Shboul & Alteneiji, 2020	Cloud ERP in an SME context. Qualitative case study using interviews and technological, management and environmental factors.	Case-study evidence has limited generalizability.
4	Yasiukovich & Haddara, 2020	Cloud-ERP in SMEs. Systematic review of 74 articles across ERP lifecycle phases.	Finds limited empirical validation in several lifecycle phases.
5	Hilal, Gadsden & Yawney, 2022	Financial anomaly detection. Review of statistical, semi-supervised and unsupervised approaches.	Focuses on financial fraud; ENTRIX applies anomaly detection to ERP expense patterns.

5. Proposed Methodology / Plan of Work

ENTRIX will follow a phased, modular development methodology based on the project proposal. Phase 1 will finalize requirements, system architecture, database design, UI/UX and authentication. The data layer will use PostgreSQL for shared organizational records, while the application will be organized into HR, Finance, Inventory, Procurement, Project/Operations, Resource Management and Analytics modules. React.js will provide the presentation layer and Node.js/Express.js will provide REST APIs and business services. JWT authentication and role-based access control will restrict module and action permissions.

Phase 2 will implement core ERP capabilities such as employee management, roles, finance and inventory. Phase 3 will implement procurement, suppliers, purchase orders, projects and workflow automation. A representative procurement flow is: Purchase Request → Manager Approval → Finance Approval → Procurement → Supplier Selection → Purchase Order → Goods Received → Inventory Updated → Invoice. Important actions will be recorded in an audit log.

Phase 4 will introduce analytics and AI services using Python/FastAPI, Scikit-learn, Pandas and NumPy. Planned functions include demand forecasting, expense anomaly detection, project budget prediction, employee workload analytics and business insights. Phase 5 will cover security hardening, testing, optimization, Dockerization, CI/CD and AWS deployment. Redis will support caching, Amazon S3 file storage and Socket.io real-time notifications.

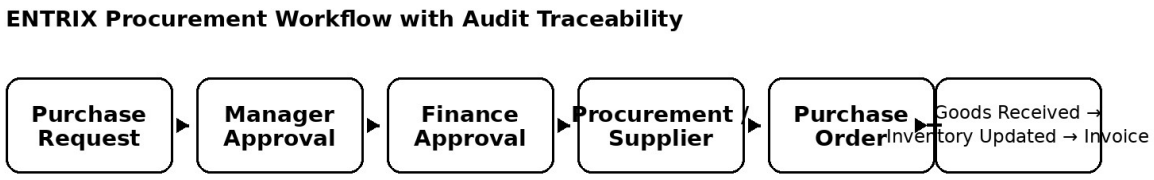


Figure 1. ENTRIX procurement workflow.

6. Technology, Tools and Platforms

Frontend: React.js, TypeScript, Tailwind CSS, Redux Toolkit, Socket.io and Vite.

Backend: Node.js, Express.js, Passport.js, JWT and Sequelize/TypeORM.

Data and storage: PostgreSQL, Redis and Amazon S3.

AI/ML: Python, FastAPI, Scikit-learn, Pandas and NumPy.

Cloud/DevOps: AWS, Docker, GitHub Actions and CloudWatch.

Testing: Jest, React Testing Library, ESLint, Prettier and SonarQube.

7. Expected Outcomes, Deliverables and Functional Specifications

The project will deliver a functional web-based ERP application with integrated organizational modules, RESTful APIs, centralized PostgreSQL storage, role-based authentication and authorization, automated approval workflows, real-time notifications, management dashboards, AI-powered predictions and anomaly detection, audit logging, automated testing and Dockerized AWS deployment. Functional specifications include employee profiles and attendance, finance records and budgets, inventory and warehouses, procurement requests and purchase orders, projects and tasks, resource utilization, operational reports and AI-supported management insights.

8. Project Scope

The project covers a modular ERP prototype for small and medium-sized organizations. It includes HR, Finance, Inventory, Procurement, Project/Operations, Resource Management and Dashboard/Analytics, together with authentication, RBAC, audit logging, notifications and selected AI features. The system is intended for academic development and demonstration. Additional business modules and third-party integrations can be considered in future versions. The current scope does not specify an industry-specific compliance package or a full production-scale enterprise rollout.

9. References

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